

Po-Wei (Brian) Chen

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EDUCATION

Soochow University, Taipei, Taiwan

B.B.A. in Financial Engineering and Actuarial Mathematics

Sept. 2022 – June 2025

- Graduated one year early (met criteria of full credit completion and top 10% class rank each semester)
- Cumulative GPA 4.0/4.0; Average 95.2/100; Class Rank 1st/53

WORK EXPERIENCE

Research Assistant (Full-time)

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Aug. 2025 – present

PI: Prof. Hsin-Cheng Huang, Distinguished Research Fellow

- Augmented diffusion-based generative models with catalytic priors that incorporate information from simpler statistical models to guide training and improve performance when only limited observations are available.
- Assisted in conference operations for the Joint Meetings of the 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference.

High School Math Tutor

Aug. 2024 – June 2025

Taiwan TACT Education (Private Tutoring Center), Minsheng Branch, Taipei, Taiwan

- Taught high school mathematics and prepared students for university entrance examinations.

HONORS & AWARDS

ISI-ISM-ISSAS Joint Conference Poster Competition Award (2026)

Title: Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors.

Outstanding Graduating Student (2025)

Soochow University, Taipei, Taiwan

Academic Excellence Award (2022–2025; awarded every semester to the top three out of a class of 50+ students)

Soochow University, Taipei, Taiwan

RESEARCH PROJECTS

Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

- Developed a novel diffusion-based framework augmented with catalytic priors that leverage simpler statistical structures to guide learning in data-scarce settings.

Empirical Asset Pricing via Machine Learning for Cryptocurrency

Soochow University, Taipei, Taiwan

- Evaluated various machine learning models for factor-based quantile investment strategies. Demonstrated the effectiveness of machine learning in cryptocurrency factor investing.

CONFERENCE PRESENTATIONS

- **“Bridging Synthetic and Real Data with Catalytic Priors for Conditional Generative Models”** 14th IASC-ARS & SoAI Joint Conference, Chiang Mai University, Chiang Mai, Thailand. December, 2026 [Accepted for Oral Presentation]
- **“Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors”** ISI-ISM-ISSAS Joint Conference, Academia Sinica, Taipei, Taiwan. February, 2026 [Poster Presentation]
- **“Empirical Asset Pricing via Machine Learning for Cryptocurrency”** Annual Conference and International Symposium of the Taiwan Risk and Insurance Association, Taipei, Taiwan. December, 2025 [Oral Presentation]

COURSEWORK & CERTIFICATIONS

Selected Coursework

Soochow University, Taipei, Taiwan

- **Mathematics & Analysis:** Advanced Calculus (textbook: William R. Wade, *An Introduction to Analysis*, 4th ed.), Calculus I–II, Linear Algebra I–II
- **Statistics:** Probability and Statistics I–II, Statistical Inference, Advanced Statistics, Regression Analysis, Applied Statistics, Introduction to AI-Aided Statistical Analysis
- **Computational & Optimization Methods:** Numerical Analysis, Operations Research, Artificial Intelligence in Finance, Computer Programming I–II, C++ Programming Language, Big Data and Fintech Practices
- **Actuarial & Financial Mathematics:** Fundamentals of Actuarial Mathematics, Mathematics of Compound Interest, Financial Calculation

Open Online Courses

- Python for Everybody Specialization, University of Michigan (via Coursera)
- Programming for Business Computing in Python (1), National Taiwan University (via Coursera)

Certification

- Actuarial Exam P (Probability), Society of Actuaries (SOA) — Passed

SKILLS & TECHNICAL

Programming: Python, R (Markdown, Shiny), C++, SQL

Machine Learning Packages: PyTorch, TensorFlow, scikit-learn

EXTRACURRICULAR ACTIVITIES

Statistics Cup Basketball Tournament Champions

Nov. 2025

Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Vice Captain of Basketball Team

2024 – 2025

Department of Financial Engineering and Actuarial Mathematics, Soochow University, Taipei, Taiwan

- Led the team to six championships in inter-departmental competitions.